A First-Principles Reconstruction of the Face-Projection Spin(10) Framework: Enumerated Six-Face Uniqueness, Automorphism-Protected Families, the Killing Contact, and the Provable Conditional Boundary

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Abstract

This note rewrites the companion lean face-projection note as a derivation from four explicitly stated principles: consistency (anomaly-free four-dimensional chiral gauge theory), face projection (the visible labels are the restriction of one complex irreducible representation of one compact simple group), a self-carried index (the protected family space is the automorphism algebra of the family curve), and a canonical contact (the Majorana contact is algebra-derived and nondegenerate). Relative to the companion note, four items move from conditional inputs or remarks to theorems. First, an exact enumeration over every simple algebra of rank 4–15 plus the exceptional groups shows that no fifteen-dimensional single-object realization of one Standard-Model family exists and that at dimension sixteen the Spin(10) half-spinor is the unique anomaly-free realization; the sixteenth state ν^c is therefore forced, not assumed. Second, the genus ladder $h^0(\Sigma_g, T_{\Sigma_g}) = 3, 1, 0$ makes three families an a-priori maximum; the Cartan criterion selects $g = 0$, so the rational family curve, the $\mathcal{O}(2)$ carrier, exactly three families, and the two-center divisor—the Poincaré–Hopf zero divisor of a Cartan flow—are derived rather than chosen. Third, the contact direction equals the Killing form of the family algebra, $B = 2\sqrt{3}\,K_{\mathrm{tr}}$, so its existence and nondegeneracy follow from the same simplicity that fixes the family number. Fourth, the hidden-messenger superpotential of the companion note's optional extension is form-unique, leaving a single complex coupling with $\lambda^2 = \zeta$. A closing boundary theorem proves, by a two-model argument with an internal witness family (the same messenger theory at variable coupling), that no value of ζ is a consequence of any such principle set: the companion note's boxed corollary is thereby upgraded from a claim boundary to a theorem. Every arithmetic and linear-algebra step is reproduced by two machine audits (44 + 24 checks); the second closes the former classical-input citations by direct computation, including the longest-Weyl-element classification of complex representations and the exactly vanishing cubic anomaly tensor of the 16 built from five fermionic modes. String-theoretic readings enter only as optional interpretations. This note adds no phenomenology; the companion note's deferred audits are unchanged.

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I. SCOPE AND RELATION TO THE COMPANION NOTE

The companion lean note[1] establishes a bounded theorem stack—the six-face Spin(10) skeleton, the $\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ family index, and the unique transvectant contact direction K_{tr}—and then deliberately stops, recording everything else as conditional inputs, deferred audits, or optional interpretations. Its central boxed statement is that face-projection motivation plus a conditional Spin(10) effective theory is *not* an unconditional principles-only proof of a grand unified theory.

This note asks the converse question: given that boundary, how much of the skeleton and of the conditional-input list can be derived from a small, explicitly stated principle set—and can the residue be proven irreducible? The answer, developed below, is:

1. the six-face object, including the sixteenth state ν^c, the hypercharge normalization $k_Y = 5/3$, and the uniqueness of the Spin(10) half-spinor, follows from the consistency and face-projection principles by exact enumeration (Sec. III);

2. the rational family curve, the $\mathcal{O}(2)$ carrier, the two-center divisor, and the family number $N_{\mathrm{fam}} = 3$ follow from the self-carried-index and canonical-contact principles, with $N_{\mathrm{fam}} \leq 3$ as an a-priori bound (Sec. IV);

3. the contact direction is the Killing form of the family algebra, with existence, nondegeneracy, and uniqueness all consequences of simplicity (Sec. V);

4. the hidden-messenger superpotential is form-unique with a single residual complex coupling (Sec. VI);

5. no value of that coupling—equivalently of ζ—is a consequence of the principles: the companion note's conditional boundary is itself a theorem (Sec. VII).

This note does not modify the companion note. It is a candidate restatement, kept in its own directory under the repository's promotion discipline: the companion note's ledger is relocated only if this reconstruction is accepted. Two machine audits replay every step (44 + 24 checks) and write ledgers with explicit negative-boundary flags (Sec. X); the provenance ledger of Sec. II A classifies every external ingredient as machine-verified, textbook-cited,

[1] `paper/gut_framework.tex` in the same repository; cited below as "the companion note."

retained input, or benchmark corroboration. The companion note's formerly deferred flavor, proton-decay, threshold, and source-sector audits have since been executed as standalone machine-checked scripts; Sec. VIII collects their outcome as a branch map and an explicit list of kill criteria, each tagged with the branch it falsifies.

Remark I.1 (Repository terminology) *In the project repository, the companion lean note, its optional hidden-messenger extension, its optional string-constrained interpretation, and the present reconstruction are filed under the working aliases "route A," "route B," "route D," and "route E," respectively. The dynamics-audit ledgers cited in Sec. VIII are filed under working aliases "DYN-0"–"DYN-9," and the three string-conditional pricing cards under "D3"–"D5." All of these aliases are file-system labels, not scientific claims; outside this remark and the literal file paths in Secs. VIII–X they do not occur in this paper.*

A. Relation to prior work

The ingredients have long histories, which the provenance ledger keeps separate from the new assembly. The Spin(10) family spinor and its binary/oscillator reading go back to Refs. [1, 2], with a modern mathematical exposition in Ref. [3]; anomaly safety of the orthogonal series is classical [4, 5], and the relevant global anomaly vanishes because $\pi_4(\mathrm{Spin}(10)) = 0$ [6]. The index-protection idea descends from the paradigm that the generation number is a compactification index [7]. The family-symmetry tradition ordinarily imposes a continuous or discrete family group externally [8]. The present reconstruction differs at three points, which we have not found combined in the literature: the family space is required to be the automorphism algebra of its own carrier (the covariance group is generated, not imposed); the family number is then bounded a priori by three, with equality forced by the Cartan criterion; and the Majorana contact direction is the Killing form of that algebra. These are claimed as new assemblies of standard mathematics, not as new mathematics.

II. THE PRINCIPLE SET AND THE ANCHORS

All derivations below are unconditional *relative to the following principles.* The principles themselves are inputs; the point of this note is to make the input list minimal and explicit, and then to prove (Sec. VII) that the remaining conditional residue cannot be absorbed into any principle set of this kind.

Principle II.1 (Consistency) *The low-energy theory is a four-dimensional chiral gauge theory; all gauge anomalies of the high-energy gauge group must cancel within the posited chiral matter content.*

Principle II.2 (Face projection) *Visible multiplet labels are not primitive. They are the irreducible summands of the restriction* $\Pi_\theta(R) = \mathrm{Res}^G_{H_\theta} R$ *of a* single *complex irreducible representation* R *of a* single *compact simple group* G *to the Standard-Model face* $H_\theta \simeq \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *[1, 2, 9, 10].*

Principle II.3 (Self-carried index) *The family multiplicity is the dimension of a protected zero-mode space of an elliptic operator on a compact internal family curve* Σ*, not a tunable integer; all other SM-charged internal sectors are absent, gapped, or vectorlike. In addition—the strengthening introduced here—the family carrier introduces no auxiliary data: it is the holomorphic tangent bundle of the family curve itself, so that the protected family space* is *the automorphism algebra of the family geometry,*

$$V = H^0(\Sigma, T_\Sigma) = \mathfrak{aut}(\Sigma).$$

Principle II.4 (Canonical contact) *The Majorana contact tensor is not adjustable data: it must be a nondegenerate invariant symmetric bilinear form on* V *determined canonically by the family geometry itself, with no externally chosen tensor.*

Two empirical anchors state what the theory is a theory of; they are shared by every model considered in this note.

Empirical Anchor II.5 (Observed content) *The Standard-Model face displays exactly the fifteen chiral Weyl states of one family,* Q, L, u^c, d^c, e^c *with their standard quantum numbers, possibly together with finitely many total SM singlets, and no exotics.*

Empirical Anchor II.6 (Existence) *At least one family exists.*

Note what is *not* anchored: the family number three is not assumed anywhere below—it is derived in Sec. IV—and no property of ζ is anchored beyond its existence as a coefficient.

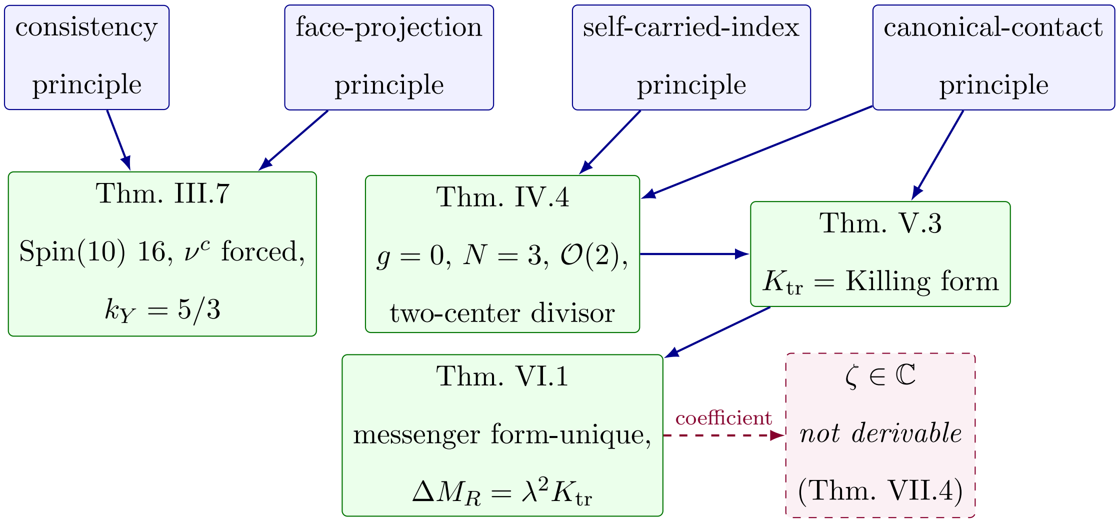

FIG. 1. Logical dependency of the reconstruction. Solid arrows are theorems; the dashed edge marks the unique point where a continuous datum enters, and Theorem VII.4 proves that no principle set of this kind can supply it.

A. Provenance and dependency ledger

Every external ingredient of this note falls into exactly one of four classes; nothing below is used outside its class.

Machine-verified here (dependency-closure audit, 24 checks).: The conjugation involutions $-w_0$ for every simple algebra of rank 4–15 and E_6, E_7, E_8, F_4; the nonzero SU(15)/SU(16) fundamental cubic traces; the exactly vanishing cubic anomaly tensor of the Spin(10) 16, built from five fermionic modes, over all 45^3 generator triples, with its Cartan weights reproducing the demicube; the exceptional minima $27, 56, 26, 248$ and the rank-16 smallest dimensions $17, 33, 32, 32$. These are the four computations listed as (i)–(iv) in Sec. III.

Machine-verified here (first-principles audit, 44 checks).: The demicube face content and $k_Y = 5/3$; the dimension-≤ 16 enumeration; the genus ladder; the one-dimensionality of the invariant contact and the identity $B = 2\sqrt{3}\,K_{\mathrm{tr}}$; the Schur complement; the ζ-independence witness.

Textbook theorems (cited with statements, not rederived).: The Weyl dimension formula and highest-weight theory [11]; Riemann–Roch and Serre duality [12]; the Cartan criterion [11]; the identification of a section's zero divisor with c_1 (Poincaré–Hopf) [12]; the Dolbeault–Hodge identification [13]; $\pi_4(\mathrm{Spin}(10)) = 0$ [6]; minimal-dimension tables for rank ≥ 17 [10].

Retained inputs (axioms and matching-level choices).: (i) the four principles; (ii) the two empirical anchors; (iii) the minimal-dimension tie-break in Theorem III.7 (an Occam clause; dropping it reopens, e.g., the E_6 27 with vectorlike extra matter); (iv) the $U(1)_R$ charge assignment of the hidden-messenger sector — the Majorana-only selection rule is *derived* from it in the companion note, but the assignment itself is an input; (v) the post-$B-L$ projector P_{ν^c}; (vi) $\mu \neq 0$ and the no-extra-unpaired-sector clause of the self-carried-index principle.

Companion-note benchmark data (corroboration only, never premises).: The ζ digits, the sensitivity windows, the $\mathbb{Z}_{N\leq 6}$ miss, the $N = 178$ diagnostic, and the phase-transfer invariants. They illustrate Theorem VII.4; no theorem in this note has them among its hypotheses.

After Sec. VII replaces the witness family by the internal hidden-messenger family, this note has *no* dependency on the repository's string-constrained sandbox: the local F-theory and instanton readings are cited only as optional UV interpretations. The item-by-item status of this closure is tracked in the roadmap file listed in Sec. X.

III. SIX FACES FROM ENUMERATION

The companion note states the six-face proposition as a "standard minimal" realization. This section upgrades it to an enumerated uniqueness theorem and shows that the sixteenth state is a prediction.

Lemma III.1 (Chirality forces complexity) *If* $\mathrm{Res}^{G}_{G_{\mathrm{SM}}} R$ *satisfies the observed-content anchor, then* R *is a complex representation of* G*.*

Proof. Restriction commutes with complex conjugation, so the restriction of a self-conjugate representation is self-conjugate. The anchored content is not self-conjugate as a G_{SM} representation: its hypercharge multiset contains $Y = 1/6$ with multiplicity six but $Y = -1/6$

with multiplicity zero, and the possible extra SM singlets are self-conjugate, so they cannot repair the mismatch. Hence R cannot be real or pseudoreal. $\square$

Lemma III.2 (Rank bound) *$G \supseteq G_{\rm SM}$ implies* $\operatorname{rank} G \geq 4$.

Proof. A maximal torus of $G_{\rm SM}$, of dimension four, embeds in a maximal torus of G. $\square$

The classification formerly rested on four classical inputs. All four are now machine-verified by the dependency-closure audit and enter as checked computations, with the textbook context retained as background [1, 4, 5, 10, 11]:

(i) the conjugation involution $-w_0$, computed from the Cartan matrix by ρ-descent for every simple algebra of rank 4–15 and for E_6, E_7, E_8, F_4, is the label reversal for A_n, the spinor-node swap for D_{2k+1}, the arm flip for E_6, and the identity ($w_0 = -1$) for B_n, C_n, D_{2k}, E_7, E_8, F_4; hence only A_n, D_{2k+1}, E_6 admit complex irreducibles;

(ii) a single Weyl fermion in the fundamental of SU(15) or SU(16) is gauge-anomalous: on an exact integer diagonal generator, $\operatorname{Tr} T^3 = 2730$ and 3360, respectively [4, 5];

(iii) the chiral 16 of Spin(10), constructed explicitly from five fermionic modes (chirality = occupation parity), satisfies $\operatorname{Tr}(\{T_a, T_b\} T_c) = 0$ exactly over all 45^3 generator triples, and its Cartan weights reproduce the demicube of Appendix A; the nonperturbative check, $\pi_4(\mathrm{Spin}(10)) = 0$, remains cited rather than computed [6];

(iv) the exceptional minima are 27 (E_6), 56 (E_7), 26 (F_4), 248 (E_8) by direct scan, so no exceptional group contributes at dimension ≤ 16.

Lemma III.3 (Monotonicity of the Weyl dimension) *For a fixed simple algebra, the Weyl dimension* $\dim \lambda = \prod_{\alpha>0} \langle \lambda + \rho, \alpha^\vee \rangle / \langle \rho, \alpha^\vee \rangle$ *is strictly increasing in each Dynkin label.*

Proof. Each factor is a positive linear form in the labels with nonnegative coefficients divided by a positive constant, and every label appears with a positive coefficient in at least one factor (e.g. the corresponding simple root). $\square$

Lemma III.3 makes the following finite search exhaustive; it is performed with exact rational arithmetic in the machine audits.

Proposition III.4 (Enumeration at dimension ≤ 16) *Among all simple compact groups of rank ≥ 4 — every algebra of rank 4–15 (A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4) is scanned directly, while rank ≥ 16 contributes nothing because the smallest rank-16 irreducibles already have dimensions $17, 33, 32, 32$ (machine-checked) and the corresponding minima grow with rank (minimality per the tables of Ref. [10], cited) — the complete list of complex irreducible representations of dimension 15 or 16 is:*

dim	*representation*	*fate*
15	$\mathrm{SU}(5): \ \mathrm{Sym}^2\, 5$	*color sextet (Lemma III.6)*
15	$\mathrm{SU}(6): \ \Lambda^2\, 6$	*content fails (Lemma III.6)*
15	$\mathrm{SU}(15):$ *fund.*	*anomalous, item (ii)*
16	$\mathrm{SU}(16):$ *fund.*	*anomalous, item (ii)*
16	$\mathrm{Spin}(10): \ 16$	*survives*

together with the conjugates of each entry.

Remark III.5 (Self-conjugate near-misses) *The same scan returns two instructive non-complex entries at these dimensions: $\mathfrak{so}(9)$ possesses a* real *spinor of dimension exactly 16, and $\mathfrak{so}(15)$ a real vector of dimension 15. Both fail Lemma III.1, not the dimension count. Complexity — that is, chirality — and not size alone is what selects* $\mathrm{Spin}(10)$.

Lemma III.6 (Content eliminations at dimension 15) *No embedding $G_{\mathrm{SM}} \subset \mathrm{SU}(5)$ or $G_{\mathrm{SM}} \subset \mathrm{SU}(6)$ makes $\mathrm{Sym}^2\, 5$ or $\Lambda^2\, 6$ satisfy the observed-content anchor.*

Proof. SU(5): if the restriction of the 5 contains no color triplet, the restriction of $\mathrm{Sym}^2 5$ contains no quarks and the anchor fails. If it contains a triplet, then $\mathrm{Sym}^2 5 \supset \mathrm{Sym}^2(3, \cdot) = (6, \cdot)$, a color sextet, which is an exotic.

SU(6): decompose the 6 under $\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$. If $6 = (3, 2)$, then $\Lambda^2 6 = (\bar{3}, 3) \oplus (6, 1)$ contains a sextet. If $6 \supset (3, 1) \oplus (\bar{3}, 1)$, then $\Lambda^2 6 \supset 3 \otimes \bar{3} \supset (8, 1)$, a color octet. If $6 = (3, 1) \oplus (1, 2) \oplus (1, 1)$, then

$$\Lambda^2 6 = (\bar{3}, 1) \oplus (3, 2) \oplus (3, 1) \oplus (1, 1) \oplus (1, 2),$$

which contains a single $\bar{3}$ and an extra $(3, 1)$; one family needs two color antitriplets (u^c, d^c) and no $(3, 1)$, so the anchor fails. If the 6 contains two triplets, $3 \otimes 3 \supset (6, 1)$ appears. No case survives. $\square$

Theorem III.7 (Enumerated six-face uniqueness) *Under the consistency and face-projection principles and the observed-content anchor, the minimal-dimension realization exists, is unique, and is the* Spin(10) *half-spinor* 16. *Its Standard-Model face is exactly*

$$\mathrm{Res}^{\mathrm{Spin}(10)}_{G_{\mathrm{SM}}} 16 = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c,$$

with multiplicities $(6, 2, 3, 3, 1, 1)$ *as Weyl states.*

Proof. By Lemma III.1 the representation is complex; by Lemma III.2 the group has rank ≥ 4, and by the computed involutions (item (i)) the complex candidates within the scan range are the A_n, D_{2k+1}, E_6 families. By the observed-content anchor the dimension is 15 plus the number of SM singlets, so the minimal candidates have dimension 15 or 16. Proposition III.4, Lemma III.6, and the exact cubic traces (item (ii)) eliminate every candidate except the Spin(10) half-spinor, which is anomaly-free by item (iii). Existence and the displayed face content are the standard Pati–Salam branching, verified at weight level in Appendix A: the sixteen even-parity D_5 half-spin weights split $8 + 8$ under the Pati–Salam face and project to exactly $(Q, L, u^c, d^c, \nu^c, e^c) = (6, 2, 3, 3, 1, 1)$. □

Corollary III.8 (The sixteenth state is a prediction) *No fifteen-dimensional single-object realization of one family exists. Under the consistency and face-projection principles the sixteenth state—a total SM singlet,* ν^c*—is forced. The essential seesaw ingredient is therefore an output of the face principle, not an input.*

Corollary III.9 (Hypercharge normalization) *On the surviving object,* $Y = T_{3R} + (B - L)/2$ *and*

$$\mathrm{Tr}\, Y^2 = \frac{10}{3}, \qquad \mathrm{Tr}\, T_{3L}^2 = 2, \qquad k_Y = \frac{5}{3},$$

exactly as in the companion note (weight-level check in Appendix A).

Remark III.10 (Face covariance in action) *The standard and flipped* $\mathrm{SU}(5) \times \mathrm{U}(1)$ *identifications of the same half-spinor produce identical multisets of SM multiplet types with the roles of* $u^c \leftrightarrow d^c$, $\nu^c \leftrightarrow e^c$ *exchanged. Two legal faces of one object carry the same content; only action-level data distinguish them. This is precisely the face-covariance statement of the companion note, realized inside Theorem III.7 rather than assumed.*

IV. THREE FAMILIES FROM THE AUTOMORPHISM PRINCIPLE

The companion note protects three families by assuming the carrier data ($C = \mathbb{CP}^1,\ R = p_+ + p_-,\ L_R \simeq \mathcal{O}(2)$) and then applying the Spin^c Dolbeault index. This section derives those data from the self-carried-index and canonical-contact principles. The analytic inputs are the standard Dolbeault–Hodge identification and Riemann–Roch with Serre duality [12, 13].

Lemma IV.1 (Genus ladder) *For a compact Riemann surface Σ_g,*

$$h^0(\Sigma_g, T_{\Sigma_g}) = \begin{cases} 3, & g = 0, \\ 1, & g = 1, \\ 0, & g \geq 2. \end{cases}$$

Proof. $T_{\Sigma_g} = K_{\Sigma_g}^{-1}$ has degree $2 - 2g$. For $g = 0$, $T \simeq \mathcal{O}(2)$: Riemann–Roch gives $h^0 - h^1 = 2 + 1 = 3$ and Serre duality gives $h^1 = h^0(K \otimes T^{-1}) = h^0(K^2) = h^0(\mathcal{O}(-4)) = 0$, so $h^0 = 3$. For $g = 1$, T is trivial and $h^0 = 1$. For $g \geq 2$, $\deg T < 0$ forces $h^0 = 0$. □

Corollary IV.2 (A-priori family bound) *Under the self-carried-index principle, the family number obeys $N_{\text{fam}} \leq 3$, with equality if and only if the family curve is rational. Three families is the maximum this protection mechanism can produce, not a coincidence within it.*

Lemma IV.3 (Cartan selection) *Among $g = 0, 1$ (the cases allowed by the existence anchor and Lemma IV.1), only $g = 0$ satisfies the canonical-contact principle.*

Proof. For $g = 1$, $\mathfrak{aut}(\Sigma_1)$ is the one-dimensional abelian translation algebra; the Killing form of an abelian Lie algebra vanishes identically ($\operatorname{ad} X = 0$), so by the Cartan criterion no canonical nondegenerate invariant form exists on V, and the canonical-contact principle fails. For $g = 0$, $V = \mathfrak{aut}(\mathbb{CP}^1) = \mathfrak{sl}_2(\mathbb{C})$ is simple, and the Cartan criterion guarantees that its Killing form is nondegenerate. □

Theorem IV.4 (Three families) *Under the self-carried-index and canonical-contact principles and the existence anchor: the family curve is $\Sigma \simeq \mathbb{CP}^1$; the family space is canonically the Möbius algebra, $V = H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq \mathfrak{sl}_2(\mathbb{C})$; the carrier is $T_{\mathbb{CP}^1} \simeq \mathcal{O}(2)$; and*

$$N_{\text{fam}} = \dim \mathfrak{sl}_2(\mathbb{C}) = 3.$$

Proof. Lemma IV.1 with the existence anchor leaves $g \in \{0,1\}$; Lemma IV.3 leaves $g = 0$. The rest is the $g = 0$ row of the ladder. □

Proposition IV.5 (The two-center divisor is Poincaré–Hopf) *Let $X_h \in V$ be any regular semisimple element (a Cartan generator, e.g. $\xi\partial_\xi$ in an affine coordinate). Then its zero divisor is a sum of two distinct points,*

$$\mathrm{div}(X_h) = p_+ + p_-, \qquad \deg \mathrm{div}(X_h) = \deg T_{\mathbb{CP}^1} = c_1(T)[\mathbb{CP}^1] = \chi(S^2) = 2,$$

and $\mathcal{O}(\mathrm{div}\, X_h) \simeq T_{\mathbb{CP}^1} \simeq \mathcal{O}(2)$. The two-center divisor $R = p_+ + p_-$ of the companion note is therefore the Euler class of the family sphere realized by a Cartan flow, and the two "centers" are the two fixed points of any Möbius flow.

Proof. $\xi\partial_\xi$ has a simple zero at $\xi = 0$; in the coordinate $\eta = 1/\xi$ one has $\xi\partial_\xi = -\eta\partial_\eta$, a simple zero at $\eta = 0$. The zero divisor of a nonzero holomorphic section of a line bundle on a curve represents its first Chern class, and $c_1(T_{S^2})[S^2] = \chi(S^2) = 2$ [12]. □

Remark IV.6 (Ledger relocation) *The companion note lists $C = \mathbb{CP}^1$, $R = p_+ + p_-$, and the protected* Spin^c $\mathcal{O}(2)$ *carrier as branch assumptions. Under the self-carried-index and canonical-contact principles all three are theorems. What the index principle retains as an assumption is the protection clause itself (no additional unpaired SM-charged sectors) and the self-carried choice; the latter does real selective work, as the next remark records.*

Remark IV.7 (Why the tangent bundle and not $\mathcal{O}(2m)$) *For $g = 0$ one has $h^0(K^{-m}) = 2m + 1$, so "some canonical bundle power" would allow any odd family number. The self-carried clause is what singles out $m = 1$: only for the tangent bundle is the family space itself the algebra that generates the family covariance (the adjoint module), rather than a module over an external symmetry. Equivalently, the $SL(2)$ bookkeeping of the companion note stops being a spurion choice and becomes the geometry of V acting on itself.*

V. THE KILLING CONTACT

The companion note proves that the invariant contact direction in $\mathrm{Sym}^2 V^\vee$ is unique (the second transvectant) and remarks that it resembles the Killing form. With Theorem IV.4 the resemblance becomes an identity, and existence, nondegeneracy, and uniqueness all follow from simplicity.

Lemma V.1 (Uniqueness) $\mathrm{Sym}^2(\mathrm{Sym}^2\mathbb{C}^2) \simeq \mathrm{Sym}^4\mathbb{C}^2 \oplus \mathrm{Sym}^0\mathbb{C}^2$, *so the trivial summand has multiplicity one and any invariant symmetric bilinear form on* V *is unique up to scale [11]. (The machine audit verifies directly that the invariance equations* $\rho(J)^T C + C\rho(J) = 0$, $J \in \{J_z, J_\pm\}$, *have a one-dimensional solution space.)*

Lemma V.2 (Existence and nondegeneracy) *The Killing form* $B(x, y) = \mathrm{Tr}(\mathrm{ad}\, x\ \mathrm{ad}\, y)$ *is an invariant symmetric bilinear form on* $V = \mathfrak{sl}_2(\mathbb{C})$, *and it is nondegenerate because* $\mathfrak{sl}_2(\mathbb{C})$ *is semisimple (Cartan criterion).*

Theorem V.3 (The contact is the Killing form) *In the spherical orthonormal family basis* (T_{+1}, T_0, T_{-1}) *with* $T_{\pm 1} = \mp J_\pm/\sqrt{2}$, $T_0 = J_z$, *where* $J_z, J_\pm$ *are the spin-1 matrices, the Killing form of the family algebra is*

$$B = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 0 \end{pmatrix} = 2\sqrt{3}\, K_{\mathrm{tr}}, \qquad K_{\mathrm{tr}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

including the sign pattern of the companion note's convention. Consequently $K_{\mathrm{tr}}^2 = \frac{1}{3}I$ *and* $K_{\mathrm{tr}}^{-1} = 3K_{\mathrm{tr}}$.

Proof. The adjoint representation of $\mathfrak{sl}_2(\mathbb{C})$ is the spin-1 representation, so $B(x, y) = \mathrm{Tr}_{3\times 3}\big(R_1(x)R_1(y)\big)$ exactly. With $\mathrm{Tr}\, J_z^2 = 2$, $\mathrm{Tr}\, J_+J_- = 4$, $\mathrm{Tr}\, J_\pm^2 = 0$:

$$B(T_0, T_0) = 2, \qquad B(T_{+1}, T_{-1}) = -\tfrac{1}{2}\,\mathrm{Tr}\, J_+J_- = -2, \qquad B(T_{\pm 1}, T_{\pm 1}) = 0,$$

which is the displayed matrix; dividing by $2\sqrt{3}$ reproduces K_{tr} with $K_{\mathrm{tr}}^2 = \frac{1}{3}I$. Uniqueness up to scale is Lemma V.1. $\square$

Remark V.4 (One principle, two consequences) *The same simplicity that selects* $g = 0$ *and hence* $N_{\mathrm{fam}} = 3$ *(Lemma IV.3) also guarantees that the seesaw contact direction exists, is unique, and is invertible. In a hypothetical* $g = 1$ *branch the Majorana contact would have no canonical direction and the messenger mass term* $K_{\mathrm{tr}}^{-1}(X, X)$ *below could not even be written. "Exactly three families" and "a working invariant seesaw contact" are two faces of one principle.*

VI. THE HIDDEN-MESSENGER MECHANISM: FORM-UNIQUENESS

The companion note's optional extension posits a sterile family messenger $X \in V^\vee$ with a Majorana-only $U(1)_R$ selection rule (kept here as an EFT matching input, together with the post-$B-L$ projector $N_i = P_{\nu^c}(16_i)$; both are tagged as retained inputs in Sec. II A). Given that rule, representation theory fixes the rest.

Theorem VI.1 (Form-uniqueness and Schur complement) *The most general superpotential quadratic in (N, X) compatible with family covariance and the Majorana-only selection rule is*

$$\frac{W_R}{M_*} = \frac{1}{2} M_V(N,N) + \lambda\, X(N) - \frac{\mu}{2} K_{\rm tr}^{-1}(X,X),$$

with $\lambda, \mu \in \mathbb{C}$ and $\mu \neq 0$ required for a well-posed messenger: the coupling term is forced to be the canonical evaluation pairing (the invariant in $V^\vee \otimes V$ is unique: the trace), and the mass term is forced onto the unique invariant line of $\mathrm{Sym}^2 V$, which is $K_{\rm tr}^{-1}$ by Lemma V.1 and Lemma V.2. After rescaling X to set $\mu = 1$, integrating out X gives exactly

$$\frac{W_R^{\rm eff}}{M_*} = \frac{1}{2}\big(M_V + \lambda^2 K_{\rm tr}\big)(N,N), \qquad \Delta M_R/M_* = \lambda^2 K_{\rm tr}.$$

The mechanism carries exactly one residual complex number, and matching to the benchmark requires $\lambda^2 = \zeta$.

Proof. The tensor structures are fixed by multiplicity one as stated; note that writing the mass term at all requires $K_{\rm tr}$ nondegenerate, i.e. Lemma V.2. The X F-term gives $\lambda N - K_{\rm tr}^{-1} X = 0$, so $X = \lambda K_{\rm tr} N$, and substitution yields $\frac{1}{2} M_V(N,N) + \lambda^2 K_{\rm tr}(N,N) - \frac{1}{2}\lambda^2 K_{\rm tr}(N,N)$, which is the displayed effective superpotential. (Replayed to 10^{-14} over random matrices in the machine audit.) □

Benchmark Datum VI.2 (Benchmark coefficient, kept as reproducibility anchor)

$$\zeta = 0.1076472949 + 0.0736514853\, i,$$

$$\lambda = \sqrt{\zeta} = 0.34502115743308964 + 0.10673473744038917\, i, \qquad |\lambda| = 0.3611535729477664.$$

As in the companion note, only the first few significant figures are physically meaningful; the digits are shared bit-identically with the verifier scripts. The tree-level matching-silence and canonical-normalization caveats of the companion note apply unchanged.

VII. THE BOUNDARY THEOREM

Everything above is a theorem of the four principles (plus the stated EFT inputs of Sec. VI). Exactly one continuous quantity remains: the value of ζ. This section proves that the remainder is not an accident of insufficient cleverness.

Lemma VII.1 (Two-model lemma) *Let $\mathcal{A}$ be any set of axioms and M_1, M_2 two models satisfying $\mathcal{A}$ that disagree about a statement D. Then neither D nor $\neg D$ is a consequence of $\mathcal{A}$.*

Proof. A consequence of $\mathcal{A}$ holds in every model of $\mathcal{A}$. □

Definition VII.2 (Admissible model class — internal witness family) *Let*

$$\mathfrak{M} = \{\, M_\lambda : \lambda \in \mathbb{C}^\times \,\}$$

be the family of post-$B-L$ effective theories of Theorem VI.1 at coupling λ, all on the same Spin(10)*/three-family skeleton. Every member satisfies the consistency principle — the messenger X is a gauge singlet, so the gauge-anomaly content is exactly that of the 16_i, which vanishes by item (iii) of Sec. III — together with the remaining three principles and both anchors, and realizes $\zeta(M_\lambda) = \lambda^2$. Hence ζ ranges over $S = \mathbb{C}^\times$. The witness family is internal: no string-theoretic input enters. (The E3-instanton reading $\zeta_\Gamma = A_\Gamma e^{2\pi i T_\Gamma}$ [14–16] realizes the same freedom as an axion-volume modulus; from here on it is cited only as an optional UV interpretation, never as a premise.)*

Lemma VII.3 (ζ-independence of the skeleton) *Every proof in Secs. III–VI is independent of the numerical value of ζ: no step references it. (The first-principles audit witnesses this by replaying the structural checks verbatim at a second value $\zeta' \neq \zeta$ far outside the benchmark windows.)*

Theorem VII.4 (The conditional boundary is necessary) *Let $\mathcal{A}$ be any axiom set all of whose axioms hold in every member of $\mathfrak{M}$ (the four principles and any ζ-blind strengthening). Then for no $\zeta_0 \in S$ is the statement $\zeta = \zeta_0$ a consequence of $\mathcal{A}$. Consequently, fixing ζ requires an axiom that fails in part of $\mathfrak{M}$—that is, new data (moduli stabilization, a hidden phase/radial sector), which is precisely a conditional input in the sense of the companion note.*

Proof. By Definition VII.2 and Lemma VII.3, pick $M_1, M_2 \in \mathfrak{M}$ with $\zeta(M_1) = \zeta_0 \neq \zeta(M_2)$; both satisfy $\mathcal{A}$. Apply Lemma VII.1 to $D:\ \zeta = \zeta_0$. $\square$

Corollary VII.5 (The companion note's boxed corollary, upgraded) *Within* $\mathfrak{M}$,

$$\boxed{\begin{array}{c} \textit{four principles} \implies \textit{the skeleton (Thms. III.7, IV.4, V.3, VI.1)}, \\ \textit{four principles} \;\not\Longrightarrow\; \zeta, \qquad \textit{hence} \qquad \textit{a principles-only unconditional GUT proof is impossible.} \end{array}}$$

The boundary drawn by the companion note is not provisional modesty; in this model class it is a theorem.

Observation VII.6 (Finite-class corroboration from the companion note's scans) *The companion note's negative results are finite-class instances of Theorem VII.4: the tested hidden D-quotients leave modulus and phase flat; small monomial/clockwork phases with denominator* $N \leq 6$ *miss* $\arg\zeta$ *by at least* 4.471595×10^{-1} rad*; the first root of unity inside the loose window* 5.424119×10^{-5} rad *appears only at* $N = 178$*; and the visible-spurion phase transfer of the companion note's finite diagnostic has no hit. All of these replay in the first-principles audit; per the provenance ledger (Sec. II A) they are benchmark corroboration and appear in no hypothesis.*

Remark VII.7 (Diophantine diagnostics are not derivations) *The approximants* $17/178$*,* $70/733$*,* $87/911$ *are ordinary continued-fraction convergents of* $\arg\zeta/2\pi$ *(verified in the audit), with phase errors* 4.15×10^{-5}, 6.68×10^{-6}, 2.73×10^{-6} rad*. Such approximants exist for any real phase; they are side diagnostics, exactly as the companion note treats them.*

Remark VII.8 (Scope honesty) *Theorem VII.4 is relative to the stated model class. A future* principled *stabilization axiom could fix* ζ*—but by the theorem it must exclude part of* $\mathfrak{M}$*, so it is a new conditional input carrying its own threshold and UV audit cost. That is the companion note's discipline restated as mathematics, not evaded.*

VIII. DYNAMICS AUDITS AND FALSIFIABILITY

The audits deferred by the companion note were executed in July 2026 as standalone machine-checked scripts, one JSON/Markdown ledger pair per audit, following the same

discipline as the audits of Sec. X (every claim gated, every negative boundary flagged). A collector script, `code/audit8_dyn8_falsifiability_collection.py`, re-verifies every number quoted in this section against its source ledger; its own ledger is `output/audit8/dyn8_falsifiability_col`
Experimental statuses are as of 2026-07-05.

A. Branch map

The results split the framework into four layers with sharply different logical statuses.

Kinematic core (supersymmetry-agnostic; machine-verified).: The theorems of Secs. III–V: the half-spinor sixteen with its forced sixteenth state, the three-family index with the a-priori bound, and the Killing-form contact direction. None of the dynamics results below touches this layer, and the $\overline{126}$-type Majorana source direction survives in every surviving branch.

Supersymmetric minimal completion (EXCLUDED as a slice family).: The benchmark supersymmetric completion (the $210 + 126 + \overline{126} + 10$ superpotential family in its minimal-spectrum slice) is excluded by the combination of gauge unification and proton decay: the unification-compatible vacuum parameter forces the unification scale down to $\sim 2 \times 10^{13}$ GeV, where the dimension-five colored-triplet channel underproduces $\tau(p \to K^+\bar{\nu})$ by 7.7 orders of magnitude against the Super-Kamiokande bound; a joint rescue scan over the vacuum parameter, the source coupling, and the supersymmetry scale finds *zero* living points, with the obstruction traced to the spectrum shape (the threshold sums force a low unification scale, a dimension-six trap). This excludes the *slice family*, not the model class; the ledgers record the distinction explicitly.

Non-supersymmetric intermediate-scale variant (ALIVE; refit required).: With supersymmetry removed the dimension-five channel is structurally absent, and two-step breaking through an intermediate gauge group decouples the intermediate scale from the unification scale. Of the four canonical chains, two survive proton decay: the left-right chain $SU(3) \times SU(2)_L \times SU(2)_R \times U(1)_{B-L}$ (intermediate scale $10^{9.4}$ GeV, unification scale $10^{16.3}$ GeV, $\tau(p \to e^+\pi^0) \sim 10^{36}$ yr) and the Pati–Salam chain $SU(4) \times SU(2)_L \times SU(2)_R$ (the chain compatible with the companion note's

210 source sector; $\tau \simeq 4.3 \times 10^{33}$ yr at one loop, within the threshold-spread band of the current bound). A perturbativity gate shows that *no* surviving chain can host the benchmark heavy Majorana tower through the renormalizable source coupling ($f < 4\pi$ fails by up to 379×): the benchmark contact card is local to the excluded supersymmetric slice.

The required non-supersymmetric re-derivation was executed as a four-stage audit series (ledgers under `output/audit9/`). Vacuum stage: the 210 was built as an explicit rank-four tensor, its full renormalizable potential constructed from a machine-tested invariant basis (the five polynomial quartic contractions span only four dimensions, and the Hodge–epsilon contraction is a genuine fifth invariant vanishing identically on the singlet vev slice), and the tree-level vacuum problem solved: the Pati–Salam vacuum is a tree-level local minimum on $\sim 16\%$ of the sampled coupling space, with every surviving scalar at or above the heavy gauge scale — so the proton-lifetime verdict is UNCHANGED by the real spectrum — while the 210-only left–right vacuum is a tree-level minimum *nowhere*, even after sweeping the epsilon coupling (the one lever that moves only off-slice masses): the left–right chain requires the adjoint (45-type) breaking route or one-loop vacuum stabilization. Source stage: the perturbative ceiling caps the neutrino Dirac scale a factor 9 (Pati–Salam) to 159 (left–right) below its SO(10)-locked value, the type-II escape falls short by 3.7–7.3 orders, and thermal leptogenesis drops below the Davidson–Ibarra floor under the rescaled renormalizable source — leaving the scale-decoupled brane-instanton source (the first pricing card below) as the only quantified surviving scenario. Contact stage: a machine-verified invariance theorem shows the normalized contact content (coefficient, direction, and contact fraction) is *exactly* invariant under a rescaling of the Dirac kernel; only the suppression scale M_* is branch-local. Consistently with Theorem VII.4, the contact coefficient is an input in every branch; only its bookkeeping moves.

String-conditional pricing cards (unpromoted).: Three cards price string-motivated escapes without deriving anything: a brane-instanton Majorana source (buys the tower *scale*, zero texture compression: the instanton data equals the full twelve-parameter Majorana data); a protection card for the hidden messenger, whose enumeration yields an abelian no-go (any charge bookkeeping that allows the Dirac Yukawa and the messenger portal ties the messenger mass to the dangerous Dirac contamination), so closure requires instanton zero-mode selectivity with a quantified gap; and a high-scale supersymmetry-breaking bridge, alive on the left-right side for a supersymmetry-breaking scale anywhere in $[2 \times 10^{10}, 3.5 \times 10^{15}]$ GeV and structurally impossible on the Pati–Salam side. These cards are conditional interpretations under the promotion discipline of the companion note's string appendix; they must not be cited as evidence.

B. Kill criteria

Each entry names the criterion, the branch it falsifies (bracketed), and the source ledger.

K1. **A fourth sequential chiral family** [kinematic core]. Falsifies the a-priori bound $N_{\mathrm{fam}} \leq 3$ of Theorem IV.4, hence the self-carried-index principle itself. Status: a fourth sequential family is strongly excluded by Higgs-coupling data; the criterion stands as the core's cleanest falsifier. (Ledger: `route_E/output/route_e_first_principles.json`.)

K2. **Dirac neutrinos** [kinematic core's physical target]. Established Dirac nature (e.g. persistent absence of neutrinoless double-beta decay together with positive evidence of lepton-number conservation) removes the Majorana contact target of Theorem V.3. Contact essentiality is posterior-wide on current oscillation data: $P(\text{contact fraction} > 0.01) = 1.000$ conditional on the benchmark Dirac kernels. (Ledger: `output/audit1/dyn4b_uncondit`

K3. **Unification × proton decay, supersymmetric slice** [supersymmetric completion — *this criterion has fired*]. $\tau(p \to K^+\bar{\nu})$ short by 7.7 orders at the unification-compatible point; zero living points in the joint rescue scan. (Ledgers: `output/audit2/dyn3_proton_` `output/audit3/dyn2b_rescue_scan.json`.)

K4. **The Pati–Salam window** [non-supersymmetric variant, 210-compatible chain]. At one loop $\tau(p \to e^+\pi^0) = 4.3 \times 10^{33}$ yr against the current bound 2.4×10^{34} yr: the chain is either already dead or sits inside the Hyper-Kamiokande discovery window — the sharpest falsifiability statement this framework currently owns. Retested with the *computed* 210 spectrum and then with the computed $\overline{126}$ spectrum at tree level (the extended-survival assumption replaced by real Hessians): the lifetime is unchanged at all sampled percentiles, so the entry now rests on three independent spectrum treatments. (Ledgers: `output/audit9/dyn9_nonsusy_intermediate.json`, `dyn9b1b_210_quartic_descent.json`, `dyn9b1c_eps_and_126_quartics.json`.)

K5. **Proton-decay channel hierarchy** [non-supersymmetric variant]. The left-right chain predicts $\tau(p \to e^+\pi^0) \sim 10^{36}$ yr (above the ten-year Hyper-Kamiokande reach; only partially probeable), and in the whole non-supersymmetric branch the gauge dimension-six channel dominates: an observed $p \to K^+\bar{\nu}$-dominated signal would disfavor the branch. Caveat from the vacuum stage: the left-right chain is kinematically realizable from the 210 but its 210-only vacuum is never a tree-level minimum (epsilon lever exhausted), so this entry's chain requires the adjoint breaking route or one-loop stabilization. (Same ledgers as K4.)

K6. **Thermal leptogenesis** [branch-tagged; soft criterion]. On the excluded supersymmetric slice, thermal unflavored leptogenesis fails by ~ 1.5 orders with the wrong central sign (success probability 0.4%; a $\times$3–10 enhancement lifts it to 34–45%). On the surviving non-supersymmetric branch the rerun with the Standard-Model loop function and Higgs normalization is a factor ~ 6 *harder* (posterior success 0/4000 unflavored; the enhancement sweet spot moves to $\times$30–100), but the gravitino/reheating ceiling dissolves by the branch itself: thermal production is unconstrained and non-thermal production opens as a qualitatively new escape, alongside flavored effects and a near-degenerate instanton-sourced tower. A soft constraint on the source scenario, not a kill. (Ledgers: `output/audit7/dyn7_leptogenesis_argzeta.json`, `output/audit9/dyn9b3_nonsusy_leptogenesis.json`.)

K7. **Messenger canonical-normalization ceilings** [hidden-messenger extension]. The one-loop audit proves light-sector silence structurally (seesaw invariance under any invertible redefinition of ν^c) and quantifies the replay ceilings: an R-breaking spurion of odd charge must satisfy $\varepsilon \lesssim 4.4 \times 10^{-4}$ against the refreshed contact-sensitivity windows; a microscopic completion violating the ceiling is excluded. (Ledger: `output/audit5/dyn5_messenger_one_loop.json`.)

K8. **Heavy-neutrino/gauge-scale coexistence** [string-conditional; UPGRADED to a branch requirement]. The instanton Majorana source *requires* the two heavier right-handed neutrinos to sit *above* the intermediate gauge scale (factors 39 and 4.8×10^3

on the Pati–Salam chain) — impossible for a renormalizable perturbative source. The source-selection audit (lock tension, type-II failure, leptogenesis floor) leaves this scale-decoupled source as the *only* quantified scenario for the alive branch with viable leptogenesis, so the coexistence prediction upgrades from an optional card to a requirement of that branch; the underlying string card itself remains unpromoted. Measuring an intermediate-scale gauge boson together with heavy-neutrino masses below it falsifies the scenario. (Ledgers: `route_d/output/d3_instanton_majorana_pricing.json`, `output/audit9/dyn9b2_nonsusy_flavor_refit.json`.)

K9. **Zero-mode selectivity gap** [string-conditional]. Closure of the non-supersymmetric protection card requires the messenger-mass instanton to miss the Dirac contamination by $\Delta S \geq 6.6$ in action (refreshed windows; 3.2 loose) at string scale 2×10^{16} GeV; any concrete embedding with a smaller gap is excluded as a completion. (Ledger: `route_d/output/d4_stueckelberg_protection.json`.)

K10. **Supersymmetry-breaking bridge** [string-conditional]. The left-right chain admits ultraviolet supersymmetry broken anywhere in $[2 \times 10^{10}, 3.5 \times 10^{15}]$ GeV (bounded below by solvability, above by the messenger matching scale, so the holomorphic matching machinery survives throughout); the Pati–Salam chain admits *no* such bridge (the anomaly-forced doubling of the breaking sector destroys every finite supersymmetric segment). If the Pati–Salam chain is established, its completion cannot be a high-scale supersymmetry bridge with 126-type content. (Ledger: `route_d/output/d5_susy_breaking_bridge.json`.)

K11. **Discrete-phase diagnostic retired** [benchmark bookkeeping]. On the refreshed Majorana card the $\mathbb{Z}_{178}$ phase candidate misses by 4.1×10^{-3} rad against the 5.4×10^{-5} rad window (76×): data supersede reproducibility anchors, and the continuous-phase reading stands. (Ledger: `output/audit1/dyn4b_unconditional_zeta.json`.)

Remark VIII.1 (What would not falsify anything) *The numerical value of the contact coefficient moving with refreshed data does not falsify the framework: by Theorem VII.4 it is an input, and its branch-locality is now a* theorem*: under a rescaling of the Dirac kernel the normalized contact content (coefficient, direction, contact fraction) is exactly invariant, and only the suppression scale M_* moves (machine-verified to 10^{-16}; ledger* *`output/audit9/dyn9b2_nonsusy_flavor_refit.json`). Likewise the exclusion in K3 is of a slice family; enlarging the Higgs content or abandoning weak-scale supersymmetry are recorded escapes, two of which (K4, K5) are now the live falsifiable branches.*

IX. REVISED STATUS LEDGER

Proved/constructed here.: Enumerated six-face uniqueness with forced ν^c and $k_Y = 5/3$ (Thm. III.7, Cors. III.8–III.9); genus ladder and the a-priori bound $N_{\rm fam} \leq 3$ (Lem. IV.1, Cor. IV.2); $g = 0$ selection, $C = \mathbb{CP}^1$, $T \simeq \mathcal{O}(2)$, $N_{\rm fam} = 3$ (Thm. IV.4); two-center divisor as Poincaré–Hopf (Prop. IV.5); contact = Killing form, with existence/nondegeneracy/uniqueness from simplicity (Thm. V.3); messenger form-uniqueness and Schur complement (Thm. VI.1); boundary necessity with the internal witness family (Def. VII.2, Thm. VII.4); the four machine-verified computations (i)–(iv) of Sec. III.

Retained inputs (items (i)–(vi) of Sec. II A).: The four principles and the two anchors; the minimal-dimension tie-break of Theorem III.7; the $U(1)_R$ charge assignment (from which the Majorana-only selection rule is derived in the companion note); the post-$B-L$ projector P_{ν^c}; $\mu \neq 0$ and the no-extra-unpaired-sector clause.

Textbook theorems and benchmark data.: Classified item by item in the provenance ledger of Sec. II A: the former are cited with statements; the latter are corroboration only and appear in no hypothesis.

Conditional data.: The value of ζ (equivalently λ or a stabilizing modulus); the benchmark M_V; the hidden phase/radial sector. By Theorem VII.4 this category cannot be emptied by principles of the stated kind.

Executed dynamics audits (July 2026; Sec. VIII).: The formerly deferred audits ran to completion along the fixed dependency graph $0 \to (1 \parallel 4a) \to 3 \to 2$: vacuum and full heavy spectrum; thresholds and unification; seesaw replay with contact posterior; $d = 5$ proton decay; leptogenesis; joint rescue scan; non-supersymmetric variant; messenger one-loop; three string-conditional pricing cards. Net outcome: supersymmetric minimal slice excluded, non-supersymmetric branch alive and falsifiable (branch map, Sec. VIII A).

Still open.: The non-supersymmetric scalar potential and spectrum, the non-supersymmetric flavor refit with contact re-extraction at the intermediate scale (*required* for the live branch), the kernel-level Dirac refit, and physical-basis proton-decay rotations.

Not claimed — and provably not claimable in $\mathfrak{M}$.: A principles-only unique GUT with predicted ζ (Cor. VII.5).

X. REPRODUCIBILITY MANIFEST

Directory names below are repository aliases (Remark I.1).

- Paper source: `route_E/tex/route_e_first_principles.tex`.
- First-principles audit (plain `python3` + `numpy`, exact rational arithmetic for the enumeration): `python3 route_E/code/verify_route_e_first_principles.py`; ledger `route_E/output/route_e_first_principles.json` and `.md` (44 checks; all pass at this writing). It replays: the demicube face content and $k_Y = 5/3$; the dimension-≤ 16 complex-irrep enumeration; the genus ladder; the one-dimensionality of the invariant-contact solution space and the identity $B = 2\sqrt{3}\,K_{\rm tr}$; the Schur complement; every number in the companion note's benchmark tables (including $\sqrt{\zeta}$, the $\mathbb{Z}_{178}$ diagnostics, and the continued-fraction convergents); and the two-model ζ-independence witness.
- Dependency-closure audit: `python3 route_E/code/verify_route_e_dependency_closure.py`; ledger `route_E/output/route_e_dependency_closure.json` and `.md` (24 checks; all pass at this writing). It verifies the four computations (i)–(iv) of Sec. III: root systems and $-w_0$ involutions for every simple algebra of rank 4–15 plus E_6, E_7, E_8, F_4; the full dimension-$\leq$ 16 scan over all those algebras with the $\mathfrak{so}(9)/\mathfrak{so}(15)$ near-misses; the exceptional minima $27, 56, 26, 248$ and rank-16 dimensions $17, 33, 32, 32$; the exact SU(15)/SU(16) cubic traces; and the exactly vanishing cubic tensor of the Fock-constructed chiral 16 with its demicube weights.
- Dependency roadmap: `route_E/ROADMAP.md` (removed, retained-and-explained, provably-not-closable, and out-of-scope items; sections E–F track the dynamics and string-conditional programs).

- Dynamics-audit ledgers (Sec. VIII): scripts `code/audit0...audit9_dyn*.py` with ledger pairs under `output/audit0..9/`; string-conditional cards `route_d/code/verify_d3..d5_*.py` with ledgers under `route_d/output/`; collector `code/audit8_dyn8_falsifiability_collection.py` with ledger `output/audit8/dyn8_falsifiability_collection.json`, which re-verifies every number in Sec. VIII against its source ledger.

- Negative-boundary flags recorded in the JSON ledgers: `zeta_value_derived: false`; `no_extra_chiral_sector_assumption_discharged: false`; `route_b_r_selection_rule_derived: false`; `companion_phenomenology_audits_replayed: false`; `witten_global_anomaly_cited_not true`; `rank_ge_17_min_dims_cited_not_computed: true`; `pslt_only_unconditional_gut_proof_c false`.

XI. CONCLUSION

This note compresses the companion note's branch assumptions into four explicit principles and pushes the theorem boundary outward to its provable maximum: the Spin(10) half-spinor with its forced sixteenth state, the rational family curve with its Poincaré–Hopf two-center divisor and exactly three automorphism-protected families, the Killing-form contact, and the form-unique hidden-messenger mechanism are all consequences; the single complex coefficient ζ provably is not. Every external ingredient carries a provenance tag—machine-verified, textbook-cited, retained input, or benchmark corroboration—the former classical inputs are computations, and the boundary theorem's witness family is internal, so string-theoretic material survives only as optional interpretation. The deepest feature of the framework is unchanged and is now sharpened: the theory contains its own boundary statement, and a complete proof of the theory therefore includes an impossibility theorem. What remains open is exactly what the companion note says remains open—the conditional data and the deferred companion audits—and Theorem VII.4 explains why that list cannot be talked away.

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Appendix A: D_5 Demicube Face Content

Let $e_1, \ldots, e_5$ be an orthonormal basis of the D_5 weight space. The positive-chirality half-spinor has the $2^{5-1} = 16$ weights $\frac{1}{2}(s_1, \ldots, s_5)$, $s_a = \pm 1$, with an even number of minus signs. Writing m_C (m_W) for the number of minus signs among $s_1 s_2 s_3$ ($s_4 s_5$), evenness of $m_C + m_W$ splits the weights into the Pati–Salam blocks

m_C, m_W parity	#weights	block
even, even	8	$(4, 2, 1)$
odd, odd	8	$(\bar{4}, 1, 2)$

with the SU(4)$_C$ convention $(m_C = 0, 1, 2, 3) \mapsto (1_{-1}, \bar{3}_{-1/3}, 3_{+1/3}, 1_{+1})$ for $(\mathrm{SU}(3)_C, B-L)$, and $T_{3L} = (s_4 + s_5)/4$, $T_{3R} = (s_4 - s_5)/4$. Applying $Y = T_{3R} + (B-L)/2$:

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$	Y	multiplicity
Q	$(3, 2)$	$1/6$	6
L	$(1, 2)$	$-1/2$	2
u^c	$(\bar{3}, 1)$	$-2/3$	3
d^c	$(\bar{3}, 1)$	$1/3$	3
ν^c	$(1, 1)$	0	1
e^c	$(1, 1)$	1	1

summing to sixteen, with exactly one total SM singlet (Cor. III.8) and

$$\mathrm{Tr}\, Y^2 = \frac{10}{3}, \qquad \mathrm{Tr}\, T_{3L}^2 = 2, \qquad k_Y = \frac{5}{3}.$$

Appendix B: Enumeration Ledger

Machine-verified complete lists over every simple algebra of rank 4–15 plus E_6, E_7, E_8, F_4 (generic Cartan-matrix engine, exact Weyl dimension formula; conjugacy tested with the involutions $-w_0$ machine-derived as item (i) of Sec. III; conjugate pairs listed once):

dim	algebra	Dynkin labels	status
16	$A_{15} = \mathrm{SU}(16)$	$(1, 0, \ldots, 0)$	anomalous (exact cubic trace)
16	$D_5 = \mathrm{Spin}(10)$	$(0, 0, 0, 0, 1)$	**survives: the** 16
15	$A_4 = \mathrm{SU}(5)$	$(2, 0, 0, 0)$	color sextet (Lem. III.6)
15	$A_5 = \mathrm{SU}(6)$	$(0, 1, 0, 0, 0)$	content fails (Lem. III.6)
15	$A_{14} = \mathrm{SU}(15)$	$(1, 0, \ldots, 0)$	anomalous (exact cubic trace)
16	$B_4 = \mathrm{Spin}(9)$	$(0, 0, 0, 1)$	self-conjugate: fails chirality
15	$B_7 = \mathrm{Spin}(15)$	$(1, 0, \ldots, 0)$	self-conjugate: fails chirality

TABLE I. Upper block: all complex irreducible representations of dimension 15 or 16 over the full scan range. Lower block: the self-conjugate near-misses of Remark III.5. E_6 starts at 27 (exceptional-minima scan); D_7 complex representations start at 64; the smallest rank-16 irreducibles have dimensions $17, 33, 32, 32$ (machine-checked), with minimality at higher rank per Ref. [10].

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